

Ameya Patil

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I design and build interactive data visualization and analysis systems to cater to the growing need of human-in-the-loop understanding of large scale network data, using an interdisciplinary approach of combining learnings from the fields of **Systems & Human Perception**. I am also interested in visualization perception research, and spreading literacy through data visualizations, especially in the space of **socio-environmental** data contexts.

EXPERIENCE

National Center for Atmospheric Research

Boulder CO, USA

Data Visualization Intern, advised by **Marlee Smith**, **Dr. Helen Kershaw** and **Dr. Moha El Gharamti**

Summer 2023

- Designed HydroVis - an interactive analysis dashboard for the WRF-Hydro hydrological forecasting model
- Implemented the dashboard backend using Python-Flask and XArray, and the renderer using D3.js
- Presented work in IEEE VIS Viz4Climate + Sustainability Workshop, 2024 as a demo paper

University of Washington

Seattle WA, USA

PhD Student, advised by **Dr. Leilani Battle**

Feb 2022 - April 2023

- Performed a design study of domain experts in whale research to gather analysis requirements
- Designed WhaleVis - an interactive analysis dashboard for modeling and analyzing historical whale hunting data as graph
- Implemented the dashboard backend using DuckDB, and the renderer using D3.js in Observable notebook
- Published work in IEEE VIS 2023 and presented the same at the International Whaling Commission Scientific Committee Meeting 2023, Bled, Slovenia and IEEE VIS 2023, Naarm, Australia

AVIZ, Inria

Saclay, Paris, France

Research Intern, advised by **Dr. Jean-Daniel Fekete**

Summer 2021

- Worked on understanding the efficacy of confidence intervals for decision making using progressive bar charts
- Proposed and studied the efficacy of two new visualization designs for progressive bar charts
- Studied the performance of humans vs automated statistical tests for answering questions based on progressive visualizations
- Published work in TVCG 2023 and presented the same at IEEE InfoVIS 2022, Oklahoma City, USA

Fraunhofer CESE

College Park MD, USA

Research Assistant Intern, advised by **Dr. Marcel Schäfer**

Summer 2019

- Worked as Java developer on the PocketSecurity project which collects data to perform user behaviour analysis
- Identified and implemented critical data probes to be collected for better analysis and improved existing probes

NVIDIA

Pune MH, India

System Software Engineer - C/C++

July 2016 - July 2018

- Worked as developer for Shadowplay - a gameplay sharing app to record, screenshot, broadcast and coplay video games
- Worked on multi-threaded and multi-processes features, GPU driver code and render pipeline
- Enhanced and monitored the automated software testing suite and guided an intern for the same

NVIDIA

Pune MH, India

Intern

July 2015 - Dec 2015

- Device Filter Drivers - C/C++: Implemented end-to-end user input redirection from input devices to a specific application using filter drivers and device notifications
- Z-buffer - Python: Implemented aesthetic visual effects such as zoom burst using the depth data of images

SKILLS

- **Programming Languages:** C/C++, Java, Python, Javascript, SQL, Cypher, GSQL
- **Libraries/Frameworks:** D3.js, Matplotlib, Vega-lite, Altair, OpenCV, MPI, OpenMP
- **Soft Skills:** Collaboration, Public Speaking, Mentoring
- **Extra-Curricular:** Digital Photography, Photo and Video Editing

LANGUAGES

- **Marathi:** Native
- **Hindi:** Fluent
- **English:** Fluent

EDUCATION

University of Washington, Seattle (UW)

Ph.D. in Computer Science, Advised by **Dr. Leilani Battle**, GPA: 4.00/4.00

WA, USA

Jan 2021 - Present

- **Thesis:** Benchmarking and Designing Scalable Interactive Network Visualization and Analysis Systems
- **Relevant Coursework:** Computing for Conservation, Computer Science Education seminar, Game Design

University of Maryland, College Park (UMD)

M.S. in Computer Science, GPA: 3.84/4.00

MD, USA

Aug 2018 - Dec 2020

- **Relevant Coursework:** Machine Learning, Geometric Computer Vision, Advanced Computer Graphics, Physically Based Modelling, Simulation & Animation, Interactive Data Analytics, Computational Geometry, Interactive Technologies in HCI, Database System Architecture and Implementation

Birla Institute of Technology and Science - Pilani (BITS)

B.E. (Honors) in Computer Science, GPA: 8.24/10.00

Goa, India

Aug 2012 - May 2016

- **Electives:** Data Mining, Data Storage Technologies and Networks, Creative Multimedia

PUBLICATIONS

1. Z. Dong, T. Barrett, **A. Patil**, Y.Shoda, L. Battle, E. Wall, "A Design Space of Behavior Change Interventions for Responsible Data Science", Proceedings of the Conference on Intelligent User Interface (IUI), 2025. DOI: 10.1145/3708359.3712140. [pdf](#)
2. Z. Dong, **A. Patil**, Y.Shoda, L. Battle, E. Wall, "Behavior Matters: An Alternative Perspective on Promoting Responsible Data Science", Proceedings of the ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW), 2025. DOI: 10.1145/3710932. [arxiv:2410.17273](#)
3. **A. Patil**, M. Smith, H. Kershaw, M. El Gharamti, "Interactive Visualization of Ensemble Data Assimilation Forecasts for Freshwater Floods", IEEE VIS Viz4Climate + Sustainability Workshop Demo, 2024. [pdf](#)
4. **A. Patil**, Z. Rand, T. Branch, L. Battle, "WhaleVis: Visualizing the History of Commercial Whaling", IEEE VIS Short Papers, 2023. DOI: 10.1109/VIS54172.2023.00028. [arxiv:2308.04552](#)
5. **A. Patil**, Z. Rand, T. Branch, L. Battle, "WhaleVis: A New Visualization Tool for the IWC Catch Database", International Whaling Commission SC/69A/GDR/04, 2023. [archive.iwc.int/SC/69A/GDR/04](#)
6. **A. Patil**, G. Richer, C. Jermaine, D. Moritz, J.-D. Fekete, "Studying Early Decision Making for Progressive Bar Charts", IEEE Transactions on Visualization and Computer Graphics, 2023. DOI: 10.1109/TVCG.2022.3209426. HAL: <https://hal.science/hal-03738461/>
7. A. Aguinaldo, P.-Y. Chiang, A. Gain, **A. Patil**, K. Pearson and S. Feizi, "Compressing GANs using Knowledge Distillation", CoRR, vol. abs/1902.00159, 2019. [arXiv:1902.00159](#)